

AI PROJECT PROPOSAL

Tic-Tac-Toe (XO) AI Agent — Web-Based Application

COURSE

Artificial Intelligence

SEMESTER

Second Semester

YEAR

2026

◆ Team Members

#	Name	Role
1	Ahmed Talaat Mahmoud	Team Leader / AI Developer
2	Abdullah Saad Rajab	Web Developer / AI Developer

◆ Project Overview

This project presents the design and implementation of an AI-powered agent capable of playing the classic game of Tic-Tac-Toe (XO) against a human user through an interactive web-based interface. The agent leverages well-established artificial intelligence search algorithms to make optimal decisions, demonstrating core AI concepts in a practical and engaging application.

The project bridges the gap between theoretical AI knowledge and real-world implementation, showcasing how intelligent agents can perceive their environment, reason about possible actions, and select optimal strategies to achieve a goal.

◆ Project Objectives

- **Intelligent Agent Design:** Implement a fully autonomous AI agent that can play Tic-Tac-Toe at an optimal or near-optimal level using decision-making algorithms.
- **Algorithm Implementation:** Apply the Minimax algorithm with Alpha-Beta Pruning to enable the agent to evaluate game states and select the best possible move efficiently.
- **Web Interface Development:** Build a clean, responsive, and user-friendly web application that allows seamless interaction between the human player and the AI agent.

- **AI Concepts Demonstration:** Showcase fundamental AI topics such as game trees, state-space search, heuristics, and adversarial search in a tangible project.
- **Difficulty Levels:** Provide multiple difficulty settings (Easy, Medium, Hard) to make the application accessible and challenging for all users.

◆ **Methodology & Technical Approach**

Component	Technology / Algorithm	Description
AI Core	Minimax + Alpha-Beta Pruning	Optimal move selection via adversarial search tree
Backend	Python / Flask or Node.js	Handles game logic, agent moves, and API endpoints
Frontend	HTML5, CSS3, JavaScript	Interactive game board and responsive UI
Communication	REST API / WebSocket	Real-time state exchange between client and server

◆ **Expected Features**

Core Features	Additional Features
• Human vs. AI game mode • Human vs. Human game mode • AI vs. AI simulation mode • Three difficulty levels (Easy / Medium / Hard) • Real-time move highlighting	• Win / Draw / Loss detection and display • Game history and score tracking • Restart and reset functionality • Responsive design (mobile & desktop) • Visual indication of AI thinking process

◆ **Expected Outcomes**

- ✓ A fully functional web-based XO (Tic-Tac-Toe) game powered by an AI agent.
- ✓ Demonstrated understanding of adversarial search and game theory in AI.
- ✓ A deployable web application accessible from any modern browser without installation.
- ✓ Complete project documentation including design diagrams, algorithm explanation, and user guide.

✓ A presentation and live demo showcasing the AI agent's decision-making process.

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